

number of the other minterm by a power of 2. Therefore, two minterms can be combined if the number of the first minterm differs by a power of 2 from a second larger number in the next section down the table.

Table 3-6: Determination of prime-implicants of Example 3-13 with decimal notation

(a)	(b)	(c)
0 ✓	0, 1(1)	0, 2, 8, 10 (2, 8)
1 ✓	0, 2(2) ✓	0, 2, 8, 10 (2, 8)
2 ✓	0, 8(8) ✓	10, 11, 14, 15 (1, 4)
8 ✓	2, 10(8) ✓	10, 11, 14, 15 (1, 4)
10 ✓	8, 10(2) ✓	
11 ✓	10, 14(4) ✓	
14 ✓	11, 15(4) ✓	
15 ✓	14, 15(1) ✓	

- Inspect every two decimal numbers in adjacent sections of the table. If the number in the section below is greater than the number in the section above by a power of 2 (i.e. 1, 2, 4, 8, 16, etc.), check both numbers to show that they have been used, and write them down in column (b).
- The pair of numbers transferred to column (b) includes a third number in parentheses that designates the power of 2 by which the numbers differ.
- The number in parentheses tells us the position of the dash in binary notation. ~~the numbers in the next section below~~
- The comparison between adjacent sections in column (b) is carried out in a similar fashion except that only those terms with the same number in parentheses are compared. The pair of numbers in one section must differ by a power of 2 from the pair of numbers in the next section. In column (c), write all four decimal numbers with the two numbers in parentheses designating the positions of the dash.